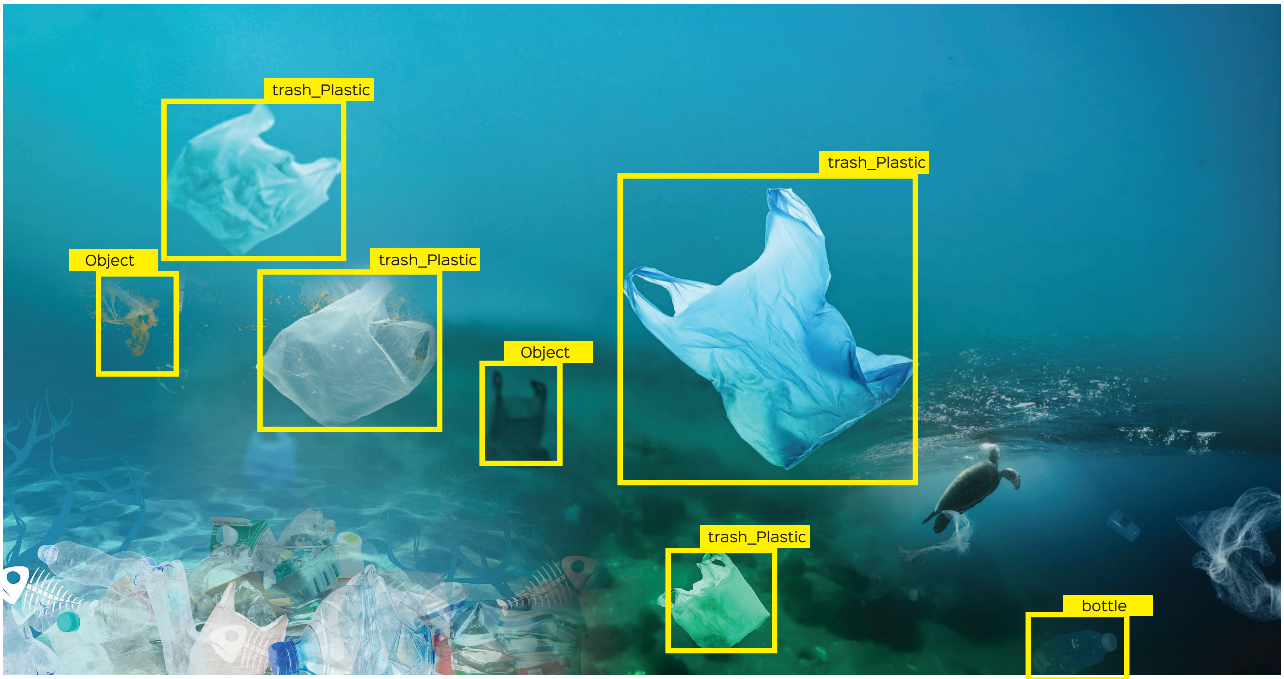
ONNX Empowers Underwater Plastic Waste Detection

The Open Neural Network Exchange (ONNX) format leverages deep learning algorithms to analyse underwater imagery and identify plastic waste in real-time. This helps to quickly prioritise areas with the highest concentration of plastic waste for clean-up efforts.



Marine pollution by plastics ranges in size from large material such as bottles and bags, down to microplastics formed from the fragmentation of plastic material. This affects the marine ecosystem deeply. Some creatures entangle in the plastic debris, while others like seabirds, turtles, fish, oysters and mussels ingest the plastics, clogging their digestive systems and causing death. Fish and birds mistake smaller plastic particles for food and feed on them, which affects the overall ecosystem too.

Currently, there are an estimated 50-75 trillion pieces of plastic and microplastics in the ocean. The plastic either ends up forming giant garbage patches or decomposes slowly into microplastics, which can enter the marine food chain and become incredibly damaging to sea life. The main source of ocean plastic pollution is land-based—80% of plastic in the ocean originates on land.

What is the Open Neural Network Exchange?

The Open Neural Network Exchange (ONNX) is an open source artificial intelligence ecosystem of tech firms and research institutions that develops open standards for describing machine learning algorithms and software tools to encourage innovation and cooperation in the AI field.

At its core, ONNX provides a standardised format for representing machine learning models. This means that different deep learning frameworks can share models seamlessly, saving valuable development time and improving model performance.

A model file that effectively depicts the link between the input data and the output predictions is the end product of a trained deep learning algorithm. Although it can be challenging to integrate a neural network into production systems, it is one of the most effective ways to create these predictive models.